

Operating System

LAB # 2

Objective

“Managing Directories , Hardware Information and kernel distribution information , Redirection and Pipes , Filters commands in Linux.”

1.Basic Commands

cat

is a simple yet powerful command that is commonly used for various file manipulation tasks.

E.g

`cat < fileName.txt / cat fileName.txt`

Stdin 0

E.g

`cat > fileName.txt`

Append >>

`cat >> fileName.txt`

Stdout 1

`cat < fileName.txt`

Stderr 2

`cd /root 2> root_error.txt` (create a new file root_error.txt and save error on it)

`hostname >> root_error.txt 2>&1` (append on root_error.txt file and save the standard output)

tac

shows the same result as cat but in reverse order.

E.g

```
tac < fileName.txt / tac fileName.txt
```

echo

is used to display text or print messages to the terminal.

E.g

```
uni="SSUET" , age=18
```

```
echo $uni  
echo $age
```

print values of the variable.

sort

is used to sort the lines of text in a file or standard input alphabetically or numerically.

E.g

```
sort fileName.txt / cat < fileName.txt | sort
```

Important Flags

- **-n** (numerically order)
- **-r** (reverse order)
- **-u** (remove duplicate lines)

Pipes

in Linux are a powerful feature that allows you to connect the output of one command to the input of another command.

Note: Single Vertical bar (| near the “enter” button on the keyboard , press shift + upward slash).

tail

shows the ending lines by using range.

E.g

`tail -5 fileName.txt` (show only last 5 lines)

grep

Helps to filter lines containing the specified pattern.

`cat < fileName.txt | grep khan`

Important Flags

- `-i` (ignore-case, ignore case distinctions in patterns and data).
- `-v` (invert-match, select non-matching lines).

wc

count no of lines ,words and also show total characters (including new line char..).

uniq

is used to filter out adjacent duplicate lines from a sorted file. It is often used in combination with the **sort** command.

Important Flags

- `-c` (counting)
- `-d` (duplicate)
- `-u` (unique)

tr

stands for "translate". It is used to translate or delete characters in a string. It is often used for converting uppercase to lowercase or replacing one character with another.

Important Flag

- -d (for deleting)

E.g

1.cat < fileName.txt | tr 'name' 'Name' (change the specific string) .

2.cat < fileName.txt | tr -d 'Hello' (delete Hello from the fileName.txt) .

3.cat < fileName.txt | tr 'a-z' 'A-Z' (change all letters from lowercase to uppercase) .

2.For Cpu Information

/proc/cpuinfo (this is a complete address)

/proc/meminfo

free

Show memory details

lspci

Lists all **Peripheral Component Interconnect (PCI)** devices connected to the system, devices that communicate with the microprocessor, such as video, audio, graphics, or network devices, and storage controllers.

dmesg

Displays the kernel ring buffer messages. It provides information about system hardware, device drivers, and kernel-level events that occur during system boot and operation.

dmidecode

Retrieves hardware information from the system's SMBIOS (System Management BIOS) table. It provides detailed information about the system's hardware components such as BIOS, motherboard, memory, CPU, and more.

4. Kernel / Distribution Information

uname

- -a (list complete info)
- -r (kernel release version)

For OS-Version

/etc/os-release

5. Basic Linux tools

find

E.g

find /etc

time

- **Real Time:** Represents the total time elapsed, including time spent waiting for resources.
- **User Time:** Represents the CPU time spent executing the command's own code.
- **System Time:** Represents the CPU time spent by the kernel on behalf of the command.

date

Shows date.

gzip

for compressing using zip algorithm

gunzip

for decompressing using zip algorithm

zcat

Show content without decompressing

bzip2

for compressing using bzip2 algorithm

bunzip2

for decompressing using bzip2 algorithm

bzcat

Show content without decompressing

Operating System

LAB # 3

Objective

“File/Directory Permission and Process Management with system call commands”

Type			
d(directory)	4 2 1 r w x r=read w=write x=execute	4 2 1 r w x r=read w=write x=execute	4 2 1 r w x r=read w=write x=execute
	4+2+1	4+0+1	000
	rwX	rw—	— — —
	owner	group	other

Permissions	Symbolic	Binary	Octal
read, write, and execute	rwX	111	7
read and write	rw-	110	6
read and execute	r-X	101	5
read	r- -	100	4
write and execute	-wX	011	3
write	-w-	010	2
execute	--X	001	1
no permission	- - -	000	0

User & Group Management

Add a user

```
sudo useradd userName
```

Add a user with details in specific group

```
sudo useradd -g groupName -s /bin/bash -c "comments" -m -d /home/userName userName
```

Add a user in multiple groups

```
usermod -G groupName userName
```

Change default group

```
usermod -g groupName userName
```

Check user

```
id userName
```

Check all Users

```
less /etc/passwd OR tail -range /etc/passwd
```

Delete a user

```
sudo userdel -r userName
```

-r (remove also home directory)

Add a Group

```
sudo groupadd groupName
```

Delete a Group

```
sudo groupdel groupName
```

Check all Groups

```
less /etc/group  OR tail -range /etc/group
```

Create a password for a user

```
sudo passwd userName
```

Switch to different user

```
su - userName / su userName
```

Operating System

LAB # 4

Objective

“Introduction to vi editor, To create & edit a file”

1.Opening file vi Editor

vi fileName.txt

2.Basic Command

For Insert/Edit A Data

Press i for insert mode

Normal Mode Again

Press Esc for normal mode

Note: Normal mode is necessary for perform the below task

For Save The Data

: (colon) + w

For Exit

Note: First to save then write below command

: (colon) + q

For Exit Without Changes

: (colon) + q + !

For Exit With Changes

: (colon) + w + q

For Undo

u

For Delete A Line

dd

For Copy A Line

yy

For Paste

p

For Cut

cc